

# Ordered Harmonic Jets

Exact orientation ranks, Stieltjes commutators,  
and depth-four Hurwitz finite parts

A research continuation prepared for the ProveIt project

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## Abstract

The harmonic-compatible regular germs of ordered multiple Hurwitz zeta functions leave a concrete problem: which of their mixed coefficients are forced by one-variable identities, and which retain ordering information? We answer this question in a precisely specified formal linear quotient. At word length  $j$  and logarithmic degree  $r$ , its dimension is  $\binom{j+r-1}{r} - p_{\leq j}(r)$ . For the complete set of coefficients entering a depth- $d$  transverse finite part, the dimension is  $2^{d-1} - p(d)$ . We give an equivariant description by augmentation spaces of permutation modules, a rational normal-form algorithm, and a residue argument showing that all these coordinates can be distinguished by positive rational rays. Single-letter closure of an individual ray occurs exactly when all its inner slopes are equal, relative to the stipulated stuffle laws.

At depth four the quotient is the direct sum of a two-dimensional standard permutation component and a one-dimensional sign component. We give the full finite part and recover its three orientation coordinates from three positive rays by an explicit rational inverse matrix. To evaluate the coordinates analytically, we prove an all-index, absolutely convergent Stieltjes commutator identity with its exact contact term, ordinary polylogarithmic Mellin representations, a curvature integral, and a Gamma-weighted antiderivative family. Rational Hurwitz shifts give explicit cyclotomic versions. These results resolve the formal depth-four organization question in the inspected predecessor and extend it to all depths. They do not assert numerical period independence, reduction to a smaller transcendental constant field, or historical priority for classical quasi-shuffle and Hurwitz continuation principles.

**Keywords.** Ordered multiple Hurwitz zeta; generalized Stieltjes constants; harmonic regularization; quasi-shuffle algebra; polylogarithm order derivatives; Gamma-weighted primitives.

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# 1 The continuation target and the meaning of the result

The predecessor *Ordered Hurwitz Germs at Every Depth* constructs harmonic-compatible holomorphic remainders near the all-one point. Its second further-research question asks for the permutation components of the lower-depth jets entering a depth-four finite part, and for a distinction between stuffle-forced combinations and genuinely ordered coordinates [1]. We address that specified question, then give an all-depth linear answer. The immediate objects are not merely numerical values: they are words representing finite logarithmic harmonic sums, equipped with explicit multiplication and a fixed constant-term prescription.

The distinction between three assertions will be used throughout. An identity between evaluated functions is proved by analytic or finite algebraic arguments. A dimension calculation in the universal word algebra is a statement relative to its defining laws. Neither one, by itself, proves arithmetic independence of the evaluated constants. In particular, the number  $2^{d-1} - p(d)$  below is a *rational linear quotient dimension*. It is not a transcendence degree, a count of algebraically independent periods, or a module rank after adjoining a field of one-variable constants.

Our principal conclusions are Theorem 4.2, which supplies an all-depth normal form; Theorems 5.1 and 5.2, which identify its ray coefficients and the exact single-letter-closed locus; Theorem 7.1, which gives every double logarithmic commutator; and Theorem 9.2, which evaluates the full depth-four finite part in three orientation coordinates. Theorems 10.2 and 11.1 connect those coordinates to Gamma-weighted integration and ordinary polylogarithmic Mellin integrals.

**Classical inputs and attribution.** Finite harmonic products, their quasi-shuffle structure, and the polynomial/Lyndon description are classical; Hoffman gives a general algebraic framework [4]. Laurent expansions of multiple zeta functions by harmonic products and analytic continuation are likewise established subjects [5]. The analytic regularization and its Gamma-generating diagonal used here are those of the inspected predecessor [1]; we reprove the required facts to make the normal-form and identity arguments self-contained. Standard one-variable identities follow the DLMF conventions [6, 7, 8, 9]. No exhaustive literature-priority assertion is made for the combinations of these tools developed here.

**Inspection boundary.** The manuscript entry point and selected portions of its integration, differentiation, and research-programme chapters were read [2, 3], together with the complete predecessor TeX. The incoming archives were inventoried by visible metadata, not exhaustively unpacked or audited. The repository reference observed during this work was

d838a9eec408cb7e36f5c9bcec194314446ed295.

The package records source paths and hashes in its provenance manifest. All claims of progress are relative to the explicit inspected question, not to every unpublished file in the repository or Library.

## 2 Logarithmic harmonic words and their constant terms

Fix  $a > 0$ . Unless stated otherwise, logarithms of positive real quantities are real logarithms. The convention for generalized Stieltjes constants is

$$\zeta(1+u, a) = \frac{1}{u} + g_a(u), \quad g_a(u) = \sum_{m \geq 0} \frac{(-1)^m \gamma_m(a)}{m!} u^m. \quad (2.1)$$

In particular  $\gamma_0(a) = -\psi(a)$ . A superscript on  $\zeta^{(m)}$  always means differentiation in the spectral argument; a derivative in  $a$  will be written explicitly.

A letter is a pair  $(k, m)$  with  $k \geq 1$  and  $m \geq 0$ . It represents  $f_{k,m}(x) = x^{-k} \log^m x$ . For a word  $w = ((k_1, m_1), \dots, (k_j, m_j))$ , put

$$W_w(N; a) = \sum_{N > n_1 > \dots > n_j \geq 0} \prod_{i=1}^j f_{k_i, m_i}(n_i + a), \quad W_\emptyset = 1. \quad (2.2)$$

The larger index carries the earlier letter. The finite generalized harmonic numbers used below are

$$H_N^{(k,m)}(a) = \sum_{n=0}^{N-1} \frac{\log^m(n+a)}{(n+a)^k}. \quad (2.3)$$

Thus  $H_N^{(1,0)}(1)$  is the ordinary  $N$ th harmonic number.

**Proposition 2.1** (Harmonic constant-term character). *Every word has a unique polynomial  $P_w(L; a)$  such that, with  $L_N = \log(N+a)$ ,*

$$W_w(N; a) = P_w(L_N; a) + O(N^{-1}(1 + \log N)^M) \quad (2.4)$$

for some  $M$ . Define  $\mathcal{R}_a(w) = P_w(0; a)$ . Let  $*$  be the positive-merge quasi-shuffle product, with

$$(k, m) \circ (\ell, n) = (k + \ell, m + n). \quad (2.5)$$

Then

$$\mathcal{R}_a(w * v) = \mathcal{R}_a(w)\mathcal{R}_a(v). \quad (2.6)$$

It agrees with every convergent sum in this class, and

$$\ell_{k,m}(a) := \mathcal{R}_a((k, m)) = \begin{cases} \gamma_m(a), & k = 1, \\ (-1)^m \zeta^{(m)}(k, a), & k \geq 2. \end{cases} \quad (2.7)$$

The estimates, and all fixed parameter derivatives needed below, are locally uniform for  $a$  in compact subsets of  $(0, \infty)$ .

*Proof.* Induct on length. Finite inner sums have at most logarithmic growth. If the first exponent is at least two, the outer series converges and its tail has the form of the error in (2.4). If the first letter is  $(1, m)$ , insert the tail polynomial into the exact increment

$$W_w(N+1; a) - W_w(N; a) = \frac{\log^m(N+a)}{N+a} W_{w'}(N; a).$$

The primitive in  $L$  of  $L^m P_{w'}(L; a)$  is a polynomial. The sum–integral difference of this increment is summable, since its derivative in the continuous variable is  $O(x^{-2}(1 + \log x)^{M'})$ . The inserted tail error is summable for the same reason. Its accumulated limit defines the missing constant. This proves existence and the error estimate; uniqueness follows because no nonzero polynomial in  $\log(N+a)$  tends to zero.

At finite cutoff, interlacing the two sets of indices and accounting for equalities proves the quasi-shuffle identity. Products of the error terms with a fixed logarithmic polynomial still tend to zero. Hence the polynomial asymptotics satisfy the same product identity. Evaluation at  $L = 0$  is multiplicative. The depth-one identities follow by Taylor expanding the Hurwitz tail, or directly from (2.1) and the convergent spectral derivatives at  $k \geq 2$ . The increment arguments and summable majorants remain uniform on compact positive  $a$ -intervals. Differentiation of those majorants gives the stated parameter regularity.  $\square$

We write  $Z_m = (1, m)$ , and frequently  $X = Z_0, Y = Z_1, Z = Z_2$ . Concatenation is not the product  $*$ : for instance

$$\mathcal{R}_a(Z_p Z_q) + \mathcal{R}_a(Z_q Z_p) = \gamma_p(a) \gamma_q(a) - (-1)^{p+q} \zeta^{(p+q)}(2, a). \quad (2.8)$$

This elementary distinction is essential at higher depth.

**Proposition 2.2** (Shift and differential transport). *For a nonempty word  $w = v(k, m)$ ,*

$$\mathcal{R}_a(w) - \mathcal{R}_{a+1}(w) = \frac{\log^m a}{a^k} \mathcal{R}_{a+1}(v). \quad (2.9)$$

*Define a derivation on concatenated words by the letter rule*

$$\mathcal{D}(k, m) = -k(k+1, m) + m(k+1, m-1), \quad (2.10)$$

*where a term with zero coefficient is omitted. Then*

$$\frac{d}{da} \mathcal{R}_a(w) = \mathcal{R}_a(\mathcal{D}w). \quad (2.11)$$

*Proof.* Separate the finite sum according to whether its smallest index is zero. The sum with all indices positive becomes a sum at  $a+1$  after subtracting one from every index. The two cutoff logarithms can be chosen identical:  $\log(N+a) = \log((N-1) + (a+1))$ . Constant terms give (2.9). Differentiating each summand gives (2.10). The locally uniform asymptotic construction in Proposition 2.1 permits differentiation of the polynomial asymptotic. Derivatives of  $L_N$  are  $O(N^{-1})$ , so contribute no constant term. This proves (2.11).  $\square$

### 3 Compatible ordered germs and transverse finite parts

Define

$$Z_d(s_1, \dots, s_d; a) = \sum_{n_1 > \dots > n_d \geq 0} \prod_{i=1}^d (n_i + a)^{-s_i}, \quad Z_0 = 1.$$

Initially all  $\operatorname{Re} s_i > 1$  suffice. The following construction provides the only local continuation needed here.

Put

$$K(t) = \frac{1}{e^t - 1} - \frac{1}{t}, \quad B(u, x) = \zeta(1+u, x+1) - \frac{x^{-u}}{u}. \quad (3.1)$$

The singularity at  $u = 0$  is removable, and

$$B(u, x) = \frac{1}{\Gamma(1+u)} \int_0^\infty t^u e^{-xt} K(t) dt, \quad \operatorname{Re} u > -1. \quad (3.2)$$

The integral follows by subtracting the usual Hurwitz and elementary Mellin integrals where they converge, then continuing. Since  $K(t) = -1/2 + O(t)$  at zero, scaling  $t = y/x$  shows that  $B$  and its fixed  $u$ -derivatives are  $O(x^{-1+\delta}(1+\log x)^M)$  on small compact  $u$ -sets. In particular,

$$A_d(\mathbf{u}; a) = \sum_{n_2 > \dots > n_d \geq 0} B(u_1, n_2 + a) \prod_{i=2}^d (n_i + a)^{-1-u_i} \quad (3.3)$$

converges normally in a sufficiently small neighborhood of zero, including after any fixed mixed derivative. Grouping by  $n_2$  gives a majorant  $O(n_2^{-2+\delta}(1+\log n_2)^M)$ , with  $\delta < 1$ .

**Proposition 3.1** (Compatible germ and harmonic jets). *Set  $\mathcal{H}_0 = 1$ ,  $\mathcal{H}_1(u; a) = g_a(u)$ , and, recursively,*

$$\mathcal{H}_d(\mathbf{u}; a) = A_d(\mathbf{u}; a) + \frac{\mathcal{H}_{d-1}(u_1 + u_2, u_3, \dots; a) - \mathcal{H}_{d-1}(u_2, u_3, \dots; a)}{u_1}. \quad (3.4)$$

*The quotient is removable. These functions are holomorphic near zero and, with  $U_k = u_1 + \dots + u_k$ ,*

$$Z_d(1 + u_1, \dots, 1 + u_d; a) = \sum_{j=0}^d \frac{\mathcal{H}_j(u_{d-j+1}, \dots, u_d; a)}{U_1 \cdots U_{d-j}}. \quad (3.5)$$

*Their Taylor coefficients are*

$$[u_1^{m_1} \cdots u_j^{m_j}] \mathcal{H}_j = \frac{(-1)^{m_1 + \dots + m_j}}{m_1! \cdots m_j!} \mathcal{R}_a(Z_{m_1} \cdots Z_{m_j}). \quad (3.6)$$

*Proof.* Sum the outer index first and use (3.1). This gives

$$Z_d(1 + \mathbf{u}; a) = \frac{1}{u_1} Z_{d-1}(1 + u_1 + u_2, 1 + u_3, \dots; a) + A_d(\mathbf{u}; a).$$

Insert the lower-depth instance of (3.5), then add and subtract  $\mathcal{H}_{d-1}(u_2, \dots)/u_1$ . Induction proves the formula and holomorphy, since a divided difference of a holomorphic function is holomorphic.

To identify the cutoff prescription rather than merely a formal subtraction, construct locally holomorphic functions  $\mathcal{Q}_j(\mathbf{u}; L)$  with

$$\mathcal{Q}_j(\mathbf{u}; L) = k_j(\mathbf{u}) + \int_0^L e^{-u_1 t} \mathcal{Q}_{j-1}(u_2, \dots; t) dt.$$

The increment proof of Proposition 2.1, now with small complex exponent perturbations, shows that the finite ordered sums equal  $\mathcal{Q}_j(\mathbf{u}; \log(N + a)) + O(N^{-1+\delta}(1 + \log N)^M)$  locally uniformly. Indeed, the difference between a discrete increment and the integral over  $[N + a, N + a + 1]$  is  $O(N^{-2+\delta}(1 + \log N)^M)$ ; so is the error from inserting the lower-depth approximation. Their normally convergent sum constructs  $k_j$ . Cauchy estimates on a slightly larger compact set justify all fixed Taylor extractions.

When all  $\operatorname{Re} u_i > 0$ , the finite sums converge. Repeated integration of the displayed recurrence gives the prefix denominators in (3.5). Uniqueness by subtraction at successive depths therefore identifies  $k_j = \mathcal{H}_j$ . Taylor extraction produces a polynomial in  $L$  whose constant is the coefficient of  $\mathcal{H}_j$ . Extracting the same coefficient from the finite sum proves (3.6).  $\square$

A transverse ray for the present purpose is a vector  $\mathbf{c} = (c_1, \dots, c_d)$  with every prefix sum  $S_k = c_1 + \dots + c_k \neq 0$ . Its finite part is the constant Laurent coefficient

$$F_d(\mathbf{c}; a) = \operatorname{FP}_{\varepsilon=0} Z_d(1 + c_1 \varepsilon, \dots, 1 + c_d \varepsilon; a).$$

The vector may have mixed signs; positive vectors are automatically admissible. No choice of approaching half-plane is needed for the local meromorphic coefficient.

**Corollary 3.2** (Exact ray coefficient formula). *For every admissible ray,*

$$F_d(\mathbf{c}; a) = \sum_{j=1}^d \frac{(-1)^{d-j}}{\prod_{k=1}^{d-j} S_k} \sum_{\substack{m_1 + \dots + m_j = d-j \\ m_i \geq 0}} \frac{\prod_{i=1}^j c_{d-j+i}^{m_i}}{\prod_{i=1}^j m_i!} \mathcal{R}_a(Z_{m_1} \cdots Z_{m_j}). \quad (3.7)$$

*Proof.* Each denominator in (3.5) is homogeneous of degree  $d - j$  in  $\varepsilon$ . Extract exactly that Taylor degree from its numerator and use (3.6). The  $j = 0$  term is a pure pole and has no constant coefficient.  $\square$

## 4 Exact formal ranks and permutation components

Let  $\mathcal{H}$  be the rational vector space freely spanned by all decorated words, with the quasi-shuffle product (2.5). Its first weight is  $\sum k_i$  and its logarithmic degree is  $\sum m_i$ . Let  $\mathcal{S}$  be its unital subalgebra generated by all individual letters. This is the *single-letter algebra*: evaluation sends it to polynomials in ordinary Stieltjes constants and spectral derivatives of Hurwitz zeta.

For fixed  $j \geq 1, r \geq 0$ , let  $V_{j,r}$  be the span of the words  $Z_{m_1} \cdots Z_{m_j}$  with  $\sum m_i = r$ . We ask for its linear quotient modulo the elements already in  $\mathcal{S}$ . This completely specifies the meaning of “orientation” in the rank theorem.

**Lemma 4.1** (Finite symmetric-sum identity). *For arbitrary  $m_1, \dots, m_j \geq 0$ , including repeated values,*

$$\sum_{\sigma \in S_j} \mathcal{R}_a(Z_{m_{\sigma(1)}} \cdots Z_{m_{\sigma(j)}}) = \sum_{\pi \in \Pi_j} (-1)^{j-|\pi|} \prod_{B \in \pi} (|B| - 1)! \ell_{|B|, \sum_{i \in B} m_i}(a). \quad (4.1)$$

*The same identity holds in  $\mathcal{H}$  before evaluation, with ordinary products on the right replaced by quasi-shuffle products. The left sum is over labelled permutations, not only distinct words.*

*Proof.* At finite cutoff, sum over  $j$  labelled indices with the condition that all are distinct. Inclusion–exclusion on their equality partitions has partition-lattice coefficient  $\prod_B (-1)^{|B|-1} (|B| - 1)!$ . This coefficient can also be verified inductively: the sum of signed cycle permutations on a set of size  $n$  is zero for  $n > 1$ , and one for  $n = 1$ . A block  $B$  of equal indices gives the single letter  $(|B|, \sum_{i \in B} m_i)$ . Ordering the distinct indices gives the labelled permutation sum. Taking constant terms proves (4.1). The same combinatorial argument interlaces formal words and proves the pre-evaluation identity.  $\square$

**Theorem 4.2** (Orientation quotient and all-depth count). *The intersection  $V_{j,r} \cap \mathcal{S}$  is exactly the span of distinct-word permutation-orbit sums. Consequently,*

$$\dim_{\mathbb{Q}} \frac{V_{j,r}}{V_{j,r} \cap \mathcal{S}} = \binom{j+r-1}{r} - p_{\leq j}(r), \quad (4.2)$$

*where  $p_{\leq j}(r)$  counts partitions of  $r$  into at most  $j$  parts, with  $p_{\leq j}(0) = 1$ . In the direct sum of all spaces entering (3.7), the quotient dimension is*

$$N_d = \dim_{\mathbb{Q}} \frac{\bigoplus_{j=1}^d V_{j,d-j}}{(\bigoplus_{j=1}^d V_{j,d-j}) \cap \mathcal{S}} = 2^{d-1} - p(d). \quad (4.3)$$

*Proof.* A quasi-shuffle product of  $h$  single letters has word length at most  $h$ . Its length- $h$  component is the full shuffle symmetrization of those letters. A monomial in single letters of first weight  $j$  has at most  $j$  factors. To have a component of length  $j$ , it must have exactly  $j$  factors, each with first exponent one. Thus the length- $j$  component of every element of  $\mathcal{S}$  of bidegree  $(j, r)$  is a linear combination of orbit sums. If that element belongs to  $V_{j,r}$ , there are no other lengths and the necessity follows. Conversely, every orbit sum belongs to  $\mathcal{S}$  by Lemma 4.1, after dividing the labelled sum by the product of the multiplicity factorials.

The number of raw words is the weak-composition count  $\binom{j+r-1}{r}$ . Their orbits are the nondecreasing  $j$ -tuples of sum  $r$ , counted by  $p_{\leq j}(r)$ . Distinct orbit sums have disjoint word supports, so are linearly independent. This proves (4.2).

The spaces with bidegrees  $(j, d-j)$  cannot mix, since  $\mathcal{S}$  is bigraded. There are  $\sum_j \binom{d-1}{j-1} = 2^{d-1}$  raw words. Adding one to every entry of a nondecreasing tuple  $(m_1, \dots, m_j)$  of sum  $d-j$  gives a partition of  $d$  into exactly  $j$  positive parts. Summing the orbit counts over  $j$  therefore gives  $p(d)$ , proving (4.3).  $\square$

**Corollary 4.3** (Equivariant description). *For each nondecreasing tuple  $\mathbf{m}$  of length  $j$  and sum  $r$ , let  $G_{\mathbf{m}} \leq S_j$  be the subgroup permuting equal entries. As an  $S_j$ -representation,*

$$\frac{V_{j,r}}{V_{j,r} \cap \mathcal{S}} \simeq \bigoplus_{\mathbf{m}} (\mathbb{Q}[S_j/G_{\mathbf{m}}]/\mathbb{Q}\mathbf{1}_{\mathbf{m}}). \quad (4.4)$$

*Equivalently each summand is the sum-zero subspace on that orbit.*

*Proof.* Each orbit is its permutation representation on the cosets of its stabilizer. Theorem 4.2 quotients out exactly its invariant orbit-sum vector. Averaging its coefficients splits off that line over  $\mathbb{Q}$ , leaving the sum-zero subspace.  $\square$

Here is a direct rational normal-form algorithm. In each orbit choose a canonical word  $w_0$ , for example the nondecreasing tuple. For every other word keep  $\delta_w = w - w_0$ . If the orbit has size  $h$  and its known orbit sum is  $T_O$ , then

$$w_0 = \frac{1}{h} \left( T_O - \sum_{w \neq w_0} \delta_w \right), \quad w = w_0 + \delta_w. \quad (4.5)$$

Lemma 4.1 evaluates  $T_O$  explicitly in single letters. Formula (4.5) is a terminating finite certificate, not a numerical relation search.

| Depth $d$ | Raw coefficients $2^{d-1}$ | Symmetric orbits $p(d)$ | $N_d$ |
|-----------|----------------------------|-------------------------|-------|
| 1         | 1                          | 1                       | 0     |
| 2         | 2                          | 2                       | 0     |
| 3         | 4                          | 3                       | 1     |
| 4         | 8                          | 5                       | 3     |
| 5         | 16                         | 7                       | 9     |
| 6         | 32                         | 11                      | 21    |
| 7         | 64                         | 15                      | 49    |
| 8         | 128                        | 22                      | 106   |
| 9         | 256                        | 30                      | 226   |
| 10        | 512                        | 42                      | 470   |
| 11        | 1024                       | 56                      | 968   |
| 12        | 2048                       | 77                      | 1971  |

**Remark 4.4** (What is not counted). The theorem does not count indecomposable Lyndon generators. Products containing an already ordered constant may relate one orientation coordinate to another and to a lower-depth ordered quantity. Nor does it compute a quotient after making all single-letter values scalars in a larger coefficient field. The first-weight/log-degree grading and the specified *rational linear* quotient are indispensable. Section 9 exhibits this distinction explicitly: a triple orientation coordinate is related to  $\gamma_0\Omega$  and to a merged-letter double sum, without ceasing to be a nonzero class in the quotient just computed.

## 5 Ray identifiability and the exact single-letter-closed locus

The quotient count is useful only if the coefficients of the rays can actually distinguish the coordinates. The nested polar denominators make this possible without numerical sampling assumptions.

**Theorem 5.1** (All raw jet coefficients are identifiable). *For fixed  $d$ , the rational functions*

$$f_{j,\mathbf{m}}(\mathbf{c}) = \frac{\prod_{i=1}^j c_{d-j+i}^{m_i}}{\prod_{k=1}^{d-j} S_k \prod_i m_i!}, \quad |\mathbf{m}| = d - j, \quad (5.1)$$

*are linearly independent over  $\mathbb{Q}$ . Consequently the  $N_d$  orientation coefficient functions in any rational normal form are also linearly independent. Finitely many positive rational rays distinguish all of them;  $N_d$  such rays can be chosen for the orientation quotient.*

*Proof.* Suppose a rational linear combination of (5.1) vanishes. Collect its terms at a fixed  $j$  into a homogeneous polynomial  $P_j(c_{d-j+1}, \dots, c_d)$  over the denominator  $S_1 \cdots S_{d-j}$ . Restrict the residue first to  $S_{d-1} = 0$ , away from the other prefix hyperplanes. Only the  $j = 1$  term can contribute, so  $P_1(c_d) = 0$ . The suffix variable  $c_d$  is unconstrained on that hyperplane, hence  $P_1$  is the zero polynomial.

Next take the residue at  $S_{d-2} = 0$ . After the  $j = 1$  term is removed, only  $j = 2$  contributes. Its two suffix variables remain arbitrary, so  $P_2 = 0$ . Continue in this fashion. The last term has no denominator and must also vanish. Within each suffix the distinct monomials are linearly independent, proving the first assertion.

A change from raw words to orbit sums and orientation differences is an invertible rational change of coordinates, so its full set of coefficient functions remains independent. Every subset, including the orientation functions, is independent. If their values on all positive rational rays spanned a proper subspace, a nonzero linear functional would annihilate all of them there. Density and continuity would give an identity on the positive open orthant, hence a rational-function identity, a contradiction. Successively choose rays whose value vectors increase the span.  $\square$

In this statement “distinguish” means that, after the single-letter contribution is subtracted, an exact finite linear system recovers the formal orientation values. It does not imply that the recovered numerical values are independent of one another or of other periods.

**Theorem 5.2** (Complete formal single-letter closure). *Let  $d \geq 3$ , and let every prefix sum of  $\mathbf{c}$  be nonzero. Replace the values in (3.7) by their universal words. That formal finite part belongs to the single-letter algebra (with the slope coefficients adjoined) if and only if*

$$c_2 = c_3 = \cdots = c_d. \quad (5.2)$$

*Proof.* The component of bidegree  $(d-1, 1)$  in (3.7) is

$$-\frac{1}{c_1} \sum_{i=1}^{d-1} c_{i+1} X^{i-1} Y X^{d-1-i}.$$

The words here constitute a single permutation orbit. By Theorem 4.2, its coefficient vector belongs to  $\mathcal{S}$  exactly when all its entries are equal. This forces (5.2). Conversely, if every inner slope is  $t$ , each suffix polynomial in (3.7) is invariant under permuting its letters: the common factor is  $t^{d-j}$  and the factorial denominator is permutation invariant. Every such combination is an orbit-sum combination, and hence lies in  $\mathcal{S}$ . The distinct bidegrees ensure that neither direction can be altered by cancellation from another layer.  $\square$

**Remark 5.3.** This is a necessary-and-sufficient result within a fixed universal law system. At particular numerical  $a$ , an extra analytic identity could reduce a ray outside (5.2). The theorem does not rule out that possibility. At  $d = 2$  there is no nontrivial orientation quotient, and the one-entry inner-slope condition is vacuous.

## 6 Gamma generation of every closed ray

The predecessor's diagonal generator provides an explicit evaluation on the complete closed locus just found. Define, as a formal power series in  $z$  with coefficients holomorphic near  $u = 0$ ,

$$\mathcal{E}_a(z, u) = \exp \left\{ zg_a(u) + \sum_{k \geq 2} \frac{(-1)^{k+1}}{k} \zeta(k + ku, a) z^k \right\}. \quad (6.1)$$

**Proposition 6.1** (Diagonal generator). *For every  $j \geq 0$ ,*

$$[z^j] \mathcal{E}_a(z, u) = \mathcal{H}_j(u, \dots, u; a), \quad \mathcal{E}_a(z, 0) = \frac{\Gamma(a)}{\Gamma(a+z)}. \quad (6.2)$$

*Proof.* For positive  $u$ , the elementary-symmetric generating identity for the convergent diagonal multiple zeta sums is

$$\sum_{d \geq 0} Z_d(1+u, \dots, 1+u; a) z^d = \exp \left\{ \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \zeta(k + ku, a) z^k \right\}.$$

The prefix denominators in (3.5) on this diagonal are  $(d-j)!u^{d-j}$ . Therefore its left side equals  $e^{z/u} \sum_{j \geq 0} \mathcal{H}_j(u, \dots, u; a) z^j$ . Cancel  $e^{z/u}$  coefficientwise and continue the holomorphic coefficients to  $u = 0$ . The Taylor expansion of  $\log \Gamma(a+z)$  then proves the second identity, since

$$\log \frac{\Gamma(a)}{\Gamma(a+z)} = \gamma_0(a)z + \sum_{k \geq 2} \frac{(-1)^{k+1}}{k} \zeta(k, a) z^k.$$

□

**Corollary 6.2** (All-depth closed-ray identity). *When  $c + rt \neq 0$  for  $0 \leq r \leq d-1$ ,*

$$F_d(c, t, \dots, t; a) = \sum_{j=1}^d \frac{t^{d-j}}{\prod_{r=0}^{d-j-1} (c + rt)} [z^j u^{d-j}] \mathcal{E}_a(z, u). \quad (6.3)$$

*Every coefficient on the right is a finite polynomial in generalized Stieltjes constants and spectral derivatives  $\zeta^{(m)}(k, a)$ ,  $k \geq 2$ . At  $t = 0$  this gives  $F_d(c, 0, \dots, 0; a) = C_d(a)$ , where*

$$C_d(a) = [z^d] \frac{\Gamma(a)}{\Gamma(a+z)} = \mathcal{R}_a(X^d). \quad (6.4)$$

*Proof.* In (3.5), the  $j$ th numerator along this ray is  $\mathcal{H}_j(t\varepsilon, \dots, t\varepsilon; a)$ . Its denominator is  $\varepsilon^{d-j} \prod_{r=0}^{d-j-1} (c + rt)$ . Use (6.2) and extract the constant coefficient. The  $t = 0$  value follows directly, or by continuity. Finite coefficient extraction from (6.1) uses only finitely many  $k$  and finitely many spectral derivatives. □

For later use, put  $x = \gamma_0(a)$  and  $z_k = \zeta(k, a)$ . Then

$$C_2 = \frac{x^2 - z_2}{2}, \quad C_3 = \frac{x^3 - 3xz_2 + 2z_3}{6}, \quad (6.5)$$

$$C_4 = \frac{x^4 - 6x^2z_2 + 3z_2^2 + 8xz_3 - 6z_4}{24}. \quad (6.6)$$

Because  $z_k = (-1)^k \psi^{(k-1)}(a)/(k-1)!$  for  $k \geq 2$ , the  $C_j$  themselves are polygamma polynomials.

## 7 Every double orientation as a convergent Stieltjes identity

Define the logarithmic commutators

$$C_{p,q}(a) = \mathcal{R}_a(Z_p Z_q) - \mathcal{R}_a(Z_q Z_p), \quad p, q \geq 0. \quad (7.1)$$

They are antisymmetric in  $p, q$ . Combined with (2.8), one commutator gives both ordered double values. The all-index contact term in the next theorem is part of the identity, not a matter of notation.

**Theorem 7.1** (All-index Stieltjes commutator). *Let  $p, q \geq 0$ ,*

$$\kappa_{p,q} = \frac{1}{q+1} - \frac{1}{p+1}, \quad n = p + q + 1.$$

*Then, for every  $a > 0$ ,*

$$C_{p,q}(a) = \kappa_{p,q} \gamma_n(a) + \sum_{v=0}^{\infty} \frac{\log^q x \gamma_p(x) - \log^p x \gamma_q(x) - \kappa_{p,q} \log^n x}{x}, \quad x = v + a. \quad (7.2)$$

*The series is ordinarily absolutely convergent, locally uniformly in  $a$ . For  $p \neq q$ , its summand has the expansion*

$$\frac{q-p}{12x^3} \log^{p+q-1} x + O\left(\frac{1 + \log^{p+q-1} x}{x^5}\right). \quad (7.3)$$

*For  $p = q$  both sides vanish identically.*

*Proof.* At depth two, (3.4) reads

$$\mathcal{H}_2(u, v) = A_2(u, v) + \frac{g_a(u+v) - g_a(v)}{u}.$$

The contribution of the divided difference to the coefficient difference corresponding to (7.1) is

$$\begin{aligned} & (-1)^{p+q} p!q! [u^p v^q - u^q v^p] \frac{g_a(u+v) - g_a(v)}{u} \\ &= -\frac{p!q!}{n!} \left\{ \binom{n}{p+1} - \binom{n}{q+1} \right\} \gamma_n(a) = \kappa_{p,q} \gamma_n(a). \end{aligned}$$

Here bracket notation denotes the difference of the two indicated coefficient extractions. This computation fixes the contact term exactly.

For  $x > 0$ , Taylor expansion of (3.1) gives

$$\partial_u^m B(0, x) = (-1)^m \left\{ \gamma_m(x+1) + \frac{\log^{m+1} x}{m+1} \right\}. \quad (7.4)$$

Differentiate the normally convergent series  $A_2(u, v) = \sum_{v \geq 0} B(u, v+a)(v+a)^{-1-v}$ . Using (7.4), its antisymmetric contribution is

$$\sum_{v \geq 0} \frac{1}{x} \{ \log^q x \gamma_p(x+1) - \log^p x \gamma_q(x+1) - \kappa_{p,q} \log^{p+q+1} x \}.$$

The Stieltjes shift  $\gamma_m(x+1) = \gamma_m(x) - \log^m x/x$  cancels between the two terms, proving (7.2).

For completeness, the first Euler–Maclaurin terms of the one-variable Hurwitz function give

$$\gamma_m(x) = -\frac{L^{m+1}}{m+1} + \frac{L^m}{2x} + \frac{L^m - mL^{m-1}}{12x^2} + O(x^{-4}(1+L^m)), \quad L = \log x. \quad (7.5)$$

The term  $mL^{m-1}$  is zero when  $m = 0$ . To obtain the sharper antisymmetric remainder in (7.3), retain the next Euler–Maclaurin term before taking the difference. It is  $x^{-4}$  times a polynomial obtained from  $(1+u)_3 e^{-uL}$ ; its top  $L^m$  coefficient is independent of  $m$ , so the top  $L^{p+q}$  term cancels. The remaining degree is at most  $p+q-1$ . The same differentiated remainder argument, or one additional Euler–Maclaurin term, proves the stated  $O$  estimate. The  $x^0$  contribution in the numerator of (7.2) is exactly cancelled by the displayed logarithmic subtraction; the  $x^{-1}$  terms cancel each other. The  $x^{-2}$  term is  $(q-p)L^{p+q-1}/12$ . Division by  $x$  proves (7.3) and absolute convergence.  $\square$

**Corollary 7.2** (Explicit ordered double normal form). *For every  $p, q \geq 0$ ,*

$$\mathcal{R}_a(Z_p Z_q) = \frac{1}{2} \{ \gamma_p(a) \gamma_q(a) - (-1)^{p+q} \zeta^{(p+q)}(2, a) + C_{p,q}(a) \}. \quad (7.6)$$

Moreover,

$$C_{p,q}(a) - C_{p,q}(a+1) = \frac{\log^q a \gamma_p(a) - \log^p a \gamma_q(a)}{a}. \quad (7.7)$$

*Proof.* Add and subtract (2.8) and (7.1). Apply (2.9) to the two ordered words for the second identity; the products of the two  $a$ -terms cancel.  $\square$

Write  $\Omega = C_{0,1}$  and  $\Omega_2 = C_{0,2}$ . The first is precisely the orientation constant in the inspected predecessor. The theorem reproduces its normalization, including the  $-\gamma_2/2$  term, and gives the next one:

$$\Omega(a) = -\frac{\gamma_2(a)}{2} + \sum_{v \geq 0} \frac{L\gamma_0(x) - \gamma_1(x) + L^2/2}{x}, \quad (7.8)$$

$$\Omega_2(a) = -\frac{2\gamma_3(a)}{3} + \sum_{v \geq 0} \frac{L^2\gamma_0(x) - \gamma_2(x) + 2L^3/3}{x}, \quad (7.9)$$

where  $x = v + a$  and  $L = \log x$ . Thus these are exact analytic coordinates, rather than residuals of an integer-relation search.

## 8 Ordinary polylogarithmic Mellin identities

For  $a, t > 0$ , define

$$\mathcal{L}_a(s; t) = \sum_{v \geq 0} \frac{e^{-(v+a)t}}{(v+a)^s} = e^{-at} \Phi(e^{-t}, s, a). \quad (8.1)$$

This is entire in  $s$  for each fixed  $t > 0$ . At  $a = 1$  it is  $\text{Li}_s(e^{-t})$ . Define the elementary reciprocal-Gamma polynomials

$$P_m(T) = \partial_u^m \frac{e^{uT}}{\Gamma(1+u)} \Big|_{u=0}. \quad (8.2)$$

With  $\gamma$  denoting Euler’s constant, the first four are

$$\begin{aligned} P_0 &= 1, & P_1 &= T + \gamma, \\ P_2 &= (T + \gamma)^2 - \zeta(2), & P_3 &= (T + \gamma)^3 - 3\zeta(2)(T + \gamma) + 2\zeta(3). \end{aligned}$$

All higher ones follow by finite coefficient extraction from the Taylor expansion of  $1/\Gamma(1+u)$ .

**Theorem 8.1** (Mellin realization of every commutator). *For  $p, q \geq 0$ , with  $n$  and  $\kappa_{p,q}$  as in Theorem 7.1,*

$$C_{p,q}(a) - \kappa_{p,q}\gamma_n(a) = (-1)^{p+q} \int_0^\infty K(t) \{P_p(\log t) \partial_s^q \mathcal{L}_a(1;t) - P_q(\log t) \partial_s^p \mathcal{L}_a(1;t)\} dt. \quad (8.3)$$

*This is an ordinary absolutely convergent integral; no finite-part interpretation is needed at either endpoint.*

*Proof.* Insert (3.2) into  $A_2$  and interchange a normally convergent sum and integral to obtain

$$A_2(u, v) = \frac{1}{\Gamma(1+u)} \int_0^\infty K(t) t^u \mathcal{L}_a(1+v; t) dt.$$

Differentiate in  $u, v$  and antisymmetrize, using (3.6) and the already computed divided-difference term. For a direct endpoint check,  $\partial_s^m \mathcal{L}_a(1; t)$  grows at most like a fixed power of  $1 + |\log t|$  as  $t \downarrow 0$ . This follows by splitting its defining sum at  $v \asymp 1/t$  and using the exponential tail. The factor  $K(t)$  is bounded there. At infinity every order derivative decays exponentially. These estimates justify all the fixed differentiations and prove the asserted convergence.  $\square$

At  $a = 1$ , (8.3) supplies an all-index family involving ordinary polylogarithms and their order derivatives. For example,

$$\Omega(1) + \frac{\gamma_2}{2} = \int_0^\infty K(t) \left\{ (\log t + \gamma) \text{Li}_1(e^{-t}) - \partial_s \text{Li}_s(e^{-t})|_{s=1} \right\} dt, \quad (8.4)$$

$$\Omega_2(1) + \frac{2\gamma_3}{3} = \int_0^\infty K(t) \left\{ \partial_s^2 \text{Li}_s(e^{-t})|_{s=1} - ((\log t + \gamma)^2 - \zeta(2)) \text{Li}_1(e^{-t}) \right\} dt. \quad (8.5)$$

Here  $\text{Li}_1(e^{-t}) = -\log(1 - e^{-t})$ . No relation between  $\Omega(1)$  and a smaller specified constant algebra is presumed.

**Proposition 8.2** (Rational shifts and cyclotomic order derivatives). *Let  $1 \leq r \leq q$  and  $\omega = e^{2\pi i/q}$ . Then*

$$\mathcal{L}_{r/q}(s; t) = q^{s-1} \sum_{h=0}^{q-1} \omega^{-rh} \text{Li}_s(\omega^h e^{-t/q}). \quad (8.6)$$

*For the ordered depth-two heat sum*

$$\mathcal{L}_a^{(2)}(s_1, s_2; t) = \sum_{\nu > \mu \geq 0} \frac{e^{-(\nu+a)t}}{(\nu+a)^{s_1} (\mu+a)^{s_2}}, \quad (8.7)$$

*one has*

$$\mathcal{L}_{r/q}^{(2)}(s_1, s_2; t) = q^{s_1+s_2-2} \sum_{h,k=0}^{q-1} \omega^{-r(h+k)} \text{Li}_{s_1, s_2}(\omega^h e^{-t/q}, \omega^k). \quad (8.8)$$

*Here  $\text{Li}_{s_1, s_2}(z_1, z_2) = \sum_{n>m \geq 1} z_1^n z_2^m / (n^{s_1} m^{s_2})$ . Both identities may be differentiated to every fixed order in the spectral variables and substituted into the ordinary integrals of this article.*

*Proof.* Write  $qv+r$  for the positive integer in the required residue class. The root-of-unity filter  $q^{-1} \sum_h \omega^{h(n-r)}$  selects that class. Applying it once proves (8.6); applying it independently to the two ordered indices proves (8.8). For  $t > 0$  the outer exponential ensures normal convergence, including every fixed logarithmic weight. In the resulting finite cyclotomic sums, the endpoint estimates in Theorem 8.1 and its depth-two analogue justify the Mellin substitutions. Derivatives of the prefactor give explicit powers of  $\log q$ .  $\square$

## 9 The complete depth-four normal form

We now answer the predecessor's depth-four question explicitly. To avoid confusing letters with numerical values, retain  $X = Z_0$ ,  $Y = Z_1$ ,  $Z = Z_2$  for words, and use

$$x = \gamma_0(a), \quad y = \gamma_1(a), \quad z = \zeta(2, a), \quad p = -\zeta'(2, a), \quad r = -\zeta'(3, a). \quad (9.1)$$

All quantities in this section are evaluated at the same positive  $a$ . Define

$$A = \mathcal{R}_a(YXX), \quad B = \mathcal{R}_a(XYX), \quad C = \mathcal{R}_a(XXY), \quad (9.2)$$

$$T = A + B + C, \quad D = A - C, \quad \Theta = A - 2B + C. \quad (9.3)$$

The final orientation coordinate is  $\Omega_2 = C_{0,2}$ .

**Proposition 9.1** (Permutation components at depth four). *Among the eight raw coefficients entering a depth-four finite part, five independent combinations are in the single-letter algebra. The remaining quotient is*

$$\text{Std}(S_3) \oplus \text{sgn}(S_2). \quad (9.4)$$

*Here the two groups act on their respective length-three and length-two word layers; this is not an assertion of a common  $S_4$  action preserving each prefix denominator. The coordinates  $(D, \Theta)$  form a rational basis of the standard component, and  $\Omega_2$  spans the sign component. The symmetric triple value is*

$$T = yC_2 - xp + r = \gamma_1(a) \frac{\gamma_0(a)^2 - \zeta(2, a)}{2} + \gamma_0(a)\zeta'(2, a) - \zeta'(3, a). \quad (9.5)$$

*Proof.* The four layers  $(j, r) = (1, 3), (2, 2), (3, 1), (4, 0)$  contain respectively 1, 3, 3, 1 words. The length-two layer consists of  $ZX, YY, XZ$ :  $ZX + XZ$  and  $YY$  are symmetric, and  $XZ - ZX$  is the sign component. The length-three layer is the permutation representation on  $YXX, XYX, XXY$ . Its sum is invariant; its sum-zero quotient is the standard two-dimensional representation. The other layers are invariant single orbits. This proves the decomposition and the count  $1 + 2 + 1 + 1 = 5$  of symmetric directions. The change of coordinates in (9.3) is invertible, with

$$A = \frac{2T + \Theta}{6} + \frac{D}{2}, \quad B = \frac{T - \Theta}{3}, \quad C = \frac{2T + \Theta}{6} - \frac{D}{2}. \quad (9.6)$$

Finally the stuffle product of  $Y$  with  $XX$  is  $YXX + XYX + XXY + (2, 1)X + X(2, 1)$ . The last pair sums to  $xp - r$ , while  $\mathcal{R}_a(XX) = C_2$ . This gives (9.5).  $\square$

Define the quadratic form

$$Q_2(u, v; a) = \frac{u^2 + v^2}{4} (x\gamma_2(a) - \zeta''(2, a)) + \frac{v^2 - u^2}{4} \Omega_2(a) + \frac{uv}{2} (y^2 - \zeta''(2, a)). \quad (9.7)$$

**Theorem 9.2** (Every ordered quartic finite part). *For every  $a > 0$  and every ray with nonzero prefix sums,*

$$F_4(c_1, c_2, c_3, c_4; a) = C_4 - \frac{1}{c_1} \left\{ \frac{c_2 + c_3 + c_4}{3} T + \frac{c_2 - c_4}{2} D + \frac{c_2 - 2c_3 + c_4}{6} \Theta \right\} + \frac{Q_2(c_3, c_4; a)}{c_1(c_1 + c_2)} - \frac{c_4^3 \gamma_3(a)}{6c_1(c_1 + c_2)(c_1 + c_2 + c_3)}. \quad (9.8)$$

*Here  $C_4$  and  $T$  are the single-letter expressions (6.6) and (9.5); the three ordered coordinates are exactly  $D, \Theta, \Omega_2$ .*

*Proof.* At  $d = 4$ , formula (3.7) is

$$F_4 = \mathcal{R}_a(X^4) - \frac{c_2 \mathcal{R}_a(YXX) + c_3 \mathcal{R}_a(XYX) + c_4 \mathcal{R}_a(XXY)}{c_1} \\ + \frac{c_3^2 \mathcal{R}_a(ZX)/2 + c_3 c_4 \mathcal{R}_a(YY) + c_4^2 \mathcal{R}_a(XZ)/2}{c_1(c_1 + c_2)} - \frac{c_4^3 \gamma_3(a)}{6c_1(c_1 + c_2)(c_1 + c_2 + c_3)}.$$

Use (9.6) in the linear term. In the quadratic term, (7.6) gives

$$\mathcal{R}_a(ZX) = \frac{x\gamma_2 - \zeta''(2, a) - \Omega_2}{2}, \quad \mathcal{R}_a(XZ) = \frac{x\gamma_2 - \zeta''(2, a) + \Omega_2}{2}, \quad \mathcal{R}_a(YY) = \frac{y^2 - \zeta''(2, a)}{2}.$$

The result is (9.7), while the constant word is  $C_4$  by (6.4).  $\square$

## 9.1 Three positive rays and an exact inverse

Let  $K_4(c; a)$  denote the right side of (9.8) with  $D, \Theta, \Omega_2$  set to zero, leaving the explicitly known single-letter part. Set

$$\Delta_1 = F_4(1, 1, 1, 2; a) - K_4(1, 1, 1, 2; a), \\ \Delta_2 = F_4(1, 1, 2, 3; a) - K_4(1, 1, 2, 3; a), \\ \Delta_3 = F_4(1, 2, 3, 4; a) - K_4(1, 2, 3, 4; a).$$

**Corollary 9.3** (Positive-ray reconstruction). *The three orientation coordinates satisfy the exact identities*

$$\begin{pmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/6 & 3/8 \\ 1 & 0 & 5/8 \\ 1 & 0 & 7/12 \end{pmatrix} \begin{pmatrix} D \\ \Theta \\ \Omega_2 \end{pmatrix}, \quad \det = -\frac{1}{144}. \quad (9.9)$$

In particular,

$$D = -14\Delta_2 + 15\Delta_3, \quad \Theta = -6\Delta_1 + 12\Delta_2 - 9\Delta_3, \quad \Omega_2 = 24\Delta_2 - 24\Delta_3. \quad (9.10)$$

*Proof.* In (9.8), the coefficients of  $(D, \Theta, \Omega_2)$  are

$$\left( -\frac{c_2 - c_4}{2c_1}, -\frac{c_2 - 2c_3 + c_4}{6c_1}, \frac{c_4^2 - c_3^2}{4c_1(c_1 + c_2)} \right).$$

Substitution of the three rays gives the displayed matrix. Rational Gaussian elimination gives its determinant and inverse.  $\square$

This corollary provides identities between genuinely ordered Laurent coefficients and convergent Stieltjes coordinates. No extrapolation from a diagonal or a fully symmetrized calculation would determine all three.

## 9.2 All equal inner slopes and a diagonal check

For  $c + rt \neq 0$  at the relevant prefixes,

$$F_4(c, t, t, t; a) = C_4 - \frac{t}{c}T + \frac{t^2}{2c(c+t)}(x\gamma_2(a) + y^2 - 2\zeta''(2, a)) \\ - \frac{t^3\gamma_3(a)}{6c(c+t)(c+2t)}. \quad (9.11)$$

This has no ordered coordinate at all, in agreement with Theorem 5.2. In particular,

$$F_4(1, 1, 1, 1; a) = C_4 - T + \frac{x\gamma_2(a) + y^2 - 2\zeta''(2, a)}{4} - \frac{\gamma_3(a)}{36}. \quad (9.12)$$

An independent exact check is available from the elementary-symmetric identity

$$\begin{aligned} Z_4(s, s, s, s; a) = \frac{1}{24} \{ & \zeta(s, a)^4 - 6\zeta(s, a)^2\zeta(2s, a) + 3\zeta(2s, a)^2 \\ & + 8\zeta(s, a)\zeta(3s, a) - 6\zeta(4s, a) \}. \end{aligned} \quad (9.13)$$

Substitute  $s = 1 + \varepsilon$  and (2.1) and extract the constant coefficient. The result is exactly (9.12); the source code performs this as a symbolic rational polynomial identity.

## 10 A Gamma-weighted differential identity and antiderivative

The first standard-component coordinate  $D$  has a useful relation to the first double commutator and to ordinarily convergent generalized harmonic sums. Define  $Q = (2, 0)$  and  $P = (2, 1)$ , and put

$$H(a) = \mathcal{R}_a(QX), \quad E(a) = \mathcal{R}_a(PX) - \mathcal{R}_a(QY). \quad (10.1)$$

For  $x_\nu = \nu + a$ ,  $L_\nu = \log x_\nu$ , let

$$h_\nu(a) = \sum_{\mu=0}^{\nu-1} \frac{1}{\mu + a}, \quad j_\nu(a) = \sum_{\mu=0}^{\nu-1} \frac{\log(\mu + a)}{\mu + a}. \quad (10.2)$$

Then

$$H(a) = \sum_{\nu \geq 0} \frac{h_\nu(a)}{x_\nu^2}, \quad E(a) = \sum_{\nu \geq 0} \frac{L_\nu h_\nu(a) - j_\nu(a)}{x_\nu^2}. \quad (10.3)$$

These sums converge absolutely and locally uniformly, including the fixed parameter derivatives used below.

**Lemma 10.1** (The merged-letter contribution). *With the notation (9.1),*

$$D = -\frac{x\Omega}{2} + \frac{xp - yz}{2} - E. \quad (10.4)$$

*Proof.* For transparency, the three finite quasi-shuffle rows needed are

$$\begin{aligned} Y * XX &= YXX + XYX + XXY + PX + XP, \\ X * YX &= 2YXX + XYX + PX + YQ, \\ X * XY &= XYX + 2XXY + QY + XP. \end{aligned}$$

Subtract the last row from the middle row and evaluate. Its left side is  $x(\mathcal{R}_a(YX) - \mathcal{R}_a(XY)) = -x\Omega$ . On the right,  $PX - QY$  contributes  $E$ , and

$$\mathcal{R}_a(YQ) - \mathcal{R}_a(XP) = (yz - r - \mathcal{R}_a(QY)) - (xp - r - \mathcal{R}_a(PX)) = yz - xp + E.$$

The triple part contributes  $2D$ . Rearrangement proves (10.4).  $\square$

This calculation explains why keeping only the highest-length part of a shuffle relation loses information. The merged letters contribute  $E$ ; that term does not disappear simply because the top-depth permutation component is two-dimensional.

**Theorem 10.2** (Gamma-weighted bridge). *For every  $a > 0$ ,*

$$\Omega'(a) = x(z - p) + yz - \zeta(3, a) - 2H(a) + 2E(a), \quad (10.5)$$

and consequently

$$2D(a) = \gamma_0(a)\zeta(2, a) - \zeta(3, a) - 2H(a) - \Gamma(a) \frac{d}{da} \left( \frac{\Omega(a)}{\Gamma(a)} \right). \quad (10.6)$$

For any positive endpoints  $\alpha, \beta$  this gives the exact antiderivative identity

$$\int_{\alpha}^{\beta} \frac{\gamma_0(a)\zeta(2, a) - \zeta(3, a) - 2H(a) - 2D(a)}{\Gamma(a)} da = \frac{\Omega(\beta)}{\Gamma(\beta)} - \frac{\Omega(\alpha)}{\Gamma(\alpha)}. \quad (10.7)$$

*Proof.* The letter derivation gives  $\mathcal{D}X = -Q$  and  $\mathcal{D}Y = -P + Q$ . Thus

$$\Omega' = -\mathcal{R}_a(QY) - \mathcal{R}_a(XP) + \mathcal{R}_a(XQ) + \mathcal{R}_a(PX) - \mathcal{R}_a(QX) + \mathcal{R}_a(YQ).$$

Use

$$\mathcal{R}_a(XP) = xp - r - \mathcal{R}_a(PX), \quad \mathcal{R}_a(YQ) = yz - r - \mathcal{R}_a(QY), \quad \mathcal{R}_a(XQ) = xz - \zeta(3, a) - H.$$

The result is (10.5). Eliminate  $E$  with (10.4) to obtain

$$2D = xz - \zeta(3, a) - 2H - x\Omega - \Omega'.$$

Since  $x = -\psi(a)$ ,  $\Gamma(a)(\Omega/\Gamma)' = \Omega' + x\Omega$ . This proves (10.6). Local uniform convergence and the parameter transport proposition ensure continuous differentiability on every compact positive interval. The fundamental theorem of calculus yields (10.7).  $\square$

At  $a = 1$  the classical Euler value  $H(1) = \zeta(3)$  makes the bridge particularly simple:

$$2D(1) = \gamma\zeta(2) - 3\zeta(3) - \gamma\Omega(1) - \Omega'(1). \quad (10.8)$$

For a direct proof of the value used here, the generating identity  $\text{Li}_{1,1}(u, 1) = \frac{1}{2} \log^2(1 - u)$  gives

$$H(1) = \frac{1}{2} \int_0^1 \frac{\log^2(1 - u)}{u} du = \frac{1}{2} \int_0^1 \frac{\log^2 v}{1 - v} dv = \zeta(3).$$

The last step is a positive termwise geometric expansion. The equality  $H(a) = \zeta(3, a)$  has not been used, and is not asserted for general  $a$ .

## 10.1 An all-index Gamma–Stieltjes antiderivative family

The first-order bridge is one member of a uniform identity. For  $q \geq 1$  define

$$D_q(a) = \mathcal{R}_a(Z_q XX) - \mathcal{R}_a(XX Z_q), \quad H_m(a) = \mathcal{R}_a((2, m)X) = \sum_{v \geq 0} \frac{\log^m(v + a) h_v(a)}{(v + a)^2}. \quad (10.9)$$

Thus  $D_1 = D$  and  $H_0 = H$ . The  $H_m$  are ordinarily convergent logarithmically weighted generalized harmonic sums.

**Theorem 10.3** (All-index Gamma bridge and unit-interval evaluation). *For  $a > 0$  and every integer  $q \geq 1$ ,*

$$\Gamma(a) \frac{d}{da} \left( \frac{C_{0,q}(a)}{\Gamma(a)} \right) = q \{ \gamma_0(a) \ell_{2,q-1}(a) - \ell_{3,q-1}(a) - 2H_{q-1}(a) \} - 2D_q(a). \quad (10.10)$$

*In particular, writing  $\gamma_q = \gamma_q(1)$ ,*

$$\int_1^2 \frac{q \{ \gamma_0(a) \ell_{2,q-1}(a) - \ell_{3,q-1}(a) - 2H_{q-1}(a) \} - 2D_q(a)}{\Gamma(a)} da = \gamma_q. \quad (10.11)$$

*The one-variable letters in these formulas are explicitly  $\ell_{k,q-1}(a) = (-1)^{q-1} \zeta^{(q-1)}(k, a)$ .*

*Proof.* Put  $Y_q = Z_q$ ,  $P_q = (2, q)$ , and  $E_q = \mathcal{R}_a(P_q X) - \mathcal{R}_a(QY_q)$ . Repeating the three stuffle rows in Lemma 10.1, with  $Y_q$  replacing  $Y$ , gives

$$2D_q = -xC_{0,q} + x\ell_{2,q} - \gamma_q(a)z - 2E_q.$$

The derivative rule  $\mathcal{D}Y_q = -P_q + qP_{q-1}$  gives

$$\begin{aligned} C'_{0,q} &= 2E_q - x\ell_{2,q} + \gamma_q(a)z + q \{ \mathcal{R}_a(XP_{q-1}) - \mathcal{R}_a(P_{q-1}X) \} \\ &= 2E_q - x\ell_{2,q} + \gamma_q(a)z + q \{ x\ell_{2,q-1} - \ell_{3,q-1} - 2H_{q-1} \}. \end{aligned}$$

Eliminate  $E_q$  and use  $x = -\psi(a)$  to obtain (10.10). This is an identity of finite quasi-shuffle expressions under the letter derivation; no convergence claim beyond the already established regularization and parameter transport is needed.

Integrate the derivative of  $C_{0,q}/\Gamma$  over  $[1, 2]$ . Since  $\Gamma(1) = \Gamma(2) = 1$ , its endpoint difference is  $C_{0,q}(2) - C_{0,q}(1)$ . At  $a = 1$ , (7.7) and  $q \geq 1$  give precisely this difference as  $\gamma_q$ . The integration is ordinary: every function involved is smooth on the compact interval.  $\square$

The case  $q = 1$  is the particularly simple identity

$$\int_1^2 \frac{\gamma_0(a) \zeta(2, a) - \zeta(3, a) - 2H(a) - 2D(a)}{\Gamma(a)} da = \gamma_1. \quad (10.12)$$

The appearance of a Stieltjes constant here comes from the exact shift of an ordered commutator, rather than from a singular endpoint or a formal exchange of divergent integrals.

## 11 The second standard component as a curvature integral

The term “curvature” here refers only to the second finite difference  $A - 2B + C$  among three placements of a logarithmic letter. It does not refer to differential geometry or an unproved sign property.

**Theorem 11.1** (Ordinary Mellin formula for the triple curvature). *Define*

$$\begin{aligned} \Xi(a) &= \int_0^\infty K(t) \{ 2\partial_{s_1} \mathcal{L}_a^{(2)}(1, 1; t) - \partial_{s_2} \mathcal{L}_a^{(2)}(1, 1; t) \\ &\quad - (\log t + \gamma) \mathcal{L}_a^{(2)}(1, 1; t) \} dt. \end{aligned} \quad (11.1)$$

*The integral is ordinarily absolutely convergent, and*

$$\Theta(a) = \Xi(a) + \frac{3}{4} \gamma_0(a) \gamma_2(a) - \frac{1}{2} \gamma_1(a)^2 - \frac{1}{4} \zeta''(2, a) - \frac{3}{4} \Omega_2(a). \quad (11.2)$$

*Proof.* Insert (3.2) into (3.3) at depth three:

$$A_3(u, v, w) = \frac{1}{\Gamma(1+u)} \int_0^\infty K(t) t^u \mathcal{L}_a^{(2)}(1+v, 1+w; t) dt.$$

Write  $\mathcal{H}_2(u, v) = \sum h_{mn} u^m v^n$ . The divided difference in (3.4) then has linear part  $h_{20}(u + 2v) + h_{11}w$ . By (3.6),

$$\Theta = -\partial_u \mathcal{H}_3(0) + 2\partial_v \mathcal{H}_3(0) - \partial_w \mathcal{H}_3(0) = \Xi + 3h_{20} - h_{11}.$$

The double normal form yields

$$h_{20} = \frac{x\gamma_2 - \zeta''(2, a) - \Omega_2}{4}, \quad h_{11} = \frac{y^2 - \zeta''(2, a)}{2}.$$

Substitution gives (11.2).

For endpoint control, the undifferentiated heat sum in (8.7), and each first order derivative, is bounded near zero by a fixed power of  $1 + |\log t|$ ; split the outer index at  $1/t$  and bound the inner harmonic sums by logarithmic polynomials. Multiplication by  $\log t$  preserves integrability. At infinity the outer exponential dominates the inner logarithmic growth. These estimates justify the integral manipulations and show that no regularized endpoint is hidden in (11.1).  $\square$

At  $a = 1$ , the heat sum is  $\text{Li}_{s_1, s_2}(e^{-t}, 1)$ ; at  $(1, 1)$  it is  $\frac{1}{2} \log^2(1 - e^{-t})$ . Consequently (11.2) is an explicit identity involving the first order derivatives of a depth-two polylogarithm, elementary logarithms, Stieltjes constants, and a Hurwitz-zeta derivative. Proposition 8.2 supplies its rational-shift cyclotomic forms.

**Proposition 11.2** (A convergent harmonic–Stieltjes series for  $\Xi$ ). *With the notation (10.2), put  $x = v + a$ ,  $L = \log x$ , and*

$$b_0(x) = \gamma_0(x + 1) + L, \quad b_1(x) = -\gamma_1(x + 1) - \frac{L^2}{2}. \quad (11.3)$$

*Then*

$$\Xi(a) = \sum_{v=0}^{\infty} \frac{(-b_1(x) - 2Lb_0(x))h_v(a) + b_0(x)j_v(a)}{x}. \quad (11.4)$$

*This is absolutely convergent. Its terms are  $O(x^{-2}(1 + \log x)^2)$ .*

*Proof.* In the normally convergent double series for  $A_3$ , the undifferentiated inner sum is  $h_v$ , while differentiation in the inner exponent gives  $-j_v$ . Also  $B(0, x) = b_0(x)$  and  $B_u(0, x) = b_1(x)$  by (7.4). Therefore  $-A_{3,u} + 2A_{3,v} - A_{3,w}$  is exactly (11.4). The bound follows from  $b_0 = O(x^{-1})$ ,  $b_1 = O(x^{-1}(1 + \log x))$  and the elementary harmonic bounds  $h_v = O(1 + \log x)$ ,  $j_v = O((1 + \log x)^2)$ .  $\square$

Together, (7.9), (10.4), (10.3), and (11.2)–(11.4) are a complete convergent-series realization of the quartic normal form. Alternatively, the Gamma bridge trades  $D$  for  $\Omega'$  and  $H$ , and the Mellin formulas trade the other coordinates for ordinary polylogarithmic integrals. These are equivalent exact representations; none presupposes an unproved reduction to a smaller constant field.

## 12 Worked identities and a comparison of representations

The results are more informative when different exact representations are kept distinct. A ray finite part, a convergent Stieltjes sum, and an ordinary Mellin integral may encode the same coordinate, but require different analytic and algebraic arguments.

## 12.1 An explicit difference of positive-ray finite parts

Let

$$F^{(2)} = F_4(1, 1, 2, 3; a), \quad F^{(3)} = F_4(1, 2, 3, 4; a).$$

Eliminating the known parts in (9.10) gives the exact identity

$$\begin{aligned} \Omega_2(a) &= 24(F^{(2)} - F^{(3)}) - 24T + 11\gamma_0(a)\gamma_2(a) + 12\gamma_1(a)^2 \\ &\quad - 23\zeta''(2, a) - \frac{13}{18}\gamma_3(a). \end{aligned} \quad (12.1)$$

For verification by hand, the two single-letter parts differ by

$$K_4(1, 1, 2, 3; a) - K_4(1, 2, 3, 4; a) = T - \frac{11}{24}(x\gamma_2 - \zeta''(2, a)) - \frac{1}{2}(y^2 - \zeta''(2, a)) + \frac{13}{432}\gamma_3.$$

Multiplying by 24 proves (12.1). Inserting (7.9) therefore yields a further convergent identity:

$$\begin{aligned} \sum_{\nu \geq 0} \frac{L^2 \gamma_0(x) - \gamma_2(x) + 2L^3/3}{x} &= 24(F^{(2)} - F^{(3)}) - 24T + 11\gamma_0(a)\gamma_2(a) + 12\gamma_1(a)^2 \\ &\quad - 23\zeta''(2, a) - \frac{1}{18}\gamma_3(a), \quad x = \nu + a, \quad L = \log x. \end{aligned} \quad (12.2)$$

This explicitly isolates the length-two sign coordinate from two rays whose length-three orientation coefficients cancel after subtraction. The equality is exact even when the individual ordered finite parts do not reduce to single-letter constants.

## 12.2 Every ordered double logarithmic moment

Combining Theorem 7.1 with Corollary 7.2 gives, without new notation,

$$\begin{aligned} \mathcal{R}_a(Z_p Z_q) &= \frac{1}{2}\gamma_p(a)\gamma_q(a) - \frac{(-1)^{p+q}}{2}\zeta^{(p+q)}(2, a) + \frac{\kappa_{p,q}}{2}\gamma_{p+q+1}(a) \\ &\quad + \frac{1}{2} \sum_{\nu \geq 0} \frac{\log^q x \gamma_p(x) - \log^p x \gamma_q(x) - \kappa_{p,q} \log^{p+q+1} x}{x}. \end{aligned} \quad (12.3)$$

Here  $x = \nu + a$ . For  $p = q$ , the final two terms vanish. For unequal indices, the sum has an  $x^{-3}$  leading tail after the cancellations built into the formula. The equality defines neither side by the other: the left side is the harmonic polynomial constant of the finite nested sum, while the right side uses ordinary one-variable Stieltjes functions in an absolutely convergent series.

## 12.3 Exact finite shifts

Iterating (7.7), for any positive integer  $N$ ,

$$C_{p,q}(a + N) = C_{p,q}(a) - \sum_{\nu=0}^{N-1} \frac{\log^q(\nu + a)\gamma_p(\nu + a) - \log^p(\nu + a)\gamma_q(\nu + a)}{\nu + a}. \quad (12.4)$$

The one-letter case of (2.9) expresses every  $\gamma_m(\nu + a)$  here as  $\gamma_m(a) - H_\nu^{(1,m)}(a)$ . Thus all positive integer translates of the commutator are determined by one base value and finite generalized harmonic sums. At  $a = 1$  and  $q > 0$ , this specializes to

$$C_{0,q}(2) - C_{0,q}(1) = \gamma_q,$$

which is exactly the endpoint mechanism behind (10.11).

## 13 Audit findings, verification, and integration

### 13.1 What is corrected, sharpened, and left open

No new mathematical error was established in the inspected predecessor’s first-commutator formula. In particular, its term  $-\gamma_2(a)/2$  is correct. The present all-index formula shows why replacing the contact term by zero, or extrapolating that first factor unchanged to all indices, would be wrong: the correct coefficient is  $\kappa_{p,q} = 1/(q+1) - 1/(p+1)$ . This is a safeguard for generalization, not an allegation that the predecessor contained that error.

The predecessor’s depth-four organization question can now be replaced by Proposition 9.1 and Theorem 9.2. Its broader all-depth version has the exact linear answer in Theorem 4.2. The words “independent coordinate” must remain qualified by the specified formal quotient. In particular, Lemma 10.1 shows that a highest-length or Lyndon-only count cannot be substituted for the complete normal form without tracking the merged-letter terms and products of earlier orientation coordinates.

These results do not settle the arithmetic reduction of  $\Omega(1)$ ,  $\Omega(1/2)$ , or  $\Omega(1/3)$  into a smaller named algebra. They do not prove the Gaussian  $S_6$  relation. The inspected research-programme chapter records an exact rejection of one particular frozen weight-nine  $S_8$  vector; that rejection is not an independence theorem and does not reject every possible relation in its basket [3]. The present work does not revisit that certificate. It also makes no claim of proof-assistant formalization.

### 13.2 Exact finite certificates

The script code `verify_exact.py` uses integer word coefficients and rational symbolic arithmetic. Its saved output records:

- 146 complete symmetric-sum expansions, including repeated logarithmic indices and every partition-lattice merge in each tested instance;
- 81 exact contact-coefficient calculations for  $0 \leq p, q \leq 8$ , and eight full word-algebra certificates of the all-index Gamma bridge;
- the rank table through  $d = 12$ , the three displayed quartic stuffle rows, the quartic linear change of coordinates, its positive-ray matrix and inverse, and the independent diagonal Newton constant-term identity.

The all-depth rank and analytic identities are not inferred from these finite checks. Their proofs are given above. The checks make the finite algebra and implementation replayable and help guard against signs, factorials, repeated-index multiplicities, and missing merged letters.

### 13.3 Numerical regressions and their limitations

The script code `verify_numeric.py` performs 83 floating-point regressions at 80 decimal working digits, using  $a = 1/2, 1, 3/2$ . It compares two cutoff/asymptotic settings,  $(N, M) = (48, 32)$  and  $(64, 40)$ ; checks single letters against separate `mpmath` Stieltjes and Hurwitz-zeta calls; compares raw harmonic constants with the convergent commutator and curvature series; and compares five quartic rays at each shift with the explicit normal form. A five-point real-parameter derivative provides a separate check of the Gamma-weighted differential bridge.

The maximum observed residual among the 82 ordinary regressions was approximately  $1.462 \times 10^{-45}$ , dominated by the comparison of truncation settings. The five-point derivative check had residual approximately

$3.477 \times 10^{-20}$  with step  $10^{-5}$ . These are observed floating-point differences, not proven error bounds. The two nested representations share the Euler–Maclaurin engine, and this shared dependency is recorded explicitly in the data. Neither tiny residuals nor stability under a larger cutoff is used as a proof of equality.

Selected diagnostic values, deliberately rounded far below the working precision, are:

| $a$ | $\Omega(a)$    | $\Omega_2(a)$   | $D(a)$          | $\Theta(a)$     |
|-----|----------------|-----------------|-----------------|-----------------|
| 1/2 | 0.013640874271 | −0.005542291305 | −2.652698961380 | 1.490647980568  |
| 1   | 0.093995347735 | 0.041077482932  | −1.272799745634 | 0.210840232622  |
| 3/2 | 0.028724389737 | 0.045438039415  | −0.951236020795 | −0.095917401069 |

The package preserves longer values, every residual, the settings, and whether each tolerance check passed. These decimals are diagnostics only; no sign or arithmetic-independence theorem is inferred from them.

### 13.4 Repository-ready contents

The delivered directory contains this article and PDF; Python verification code; JSON and text outputs; a provenance manifest; build instructions; and a short manuscript insertion fragment with namespaced labels. A suggested report location is

Analysis/Polylogarithms/docs/reports/ordered-jet-normal-forms/.

The fragment is intended for the integration/differentiation part of the manuscript, with a cross-reference from the research-programme chapter. Its status note replaces the specified formal depth-four question, while retaining the arithmetic evaluation questions. No remote repository file was modified during this work.

## 14 Further research questions

The following problems are deliberately more specific than asking for additional high-precision fits.

**1. Evaluation kernels at rational shifts.** For  $a = 1, 1/2, 1/3$ , fix an explicit comparison algebra generated by selected Gamma values, ordinary zeta derivatives, Dirichlet- $L$  derivatives, and cyclotomic polylogarithms. Determine which classes from Theorem 4.2 vanish after numerical evaluation modulo that algebra. The first concrete targets are  $\Omega_2(a)$  and the combination  $\Xi(a) - 3\Omega_2(a)/4$ . The formal rank is not an obstruction to additional analytic reductions.

**2. Linear quotients versus scalar extension.** Compute the relevant quotient after products of previously known ordered coordinates with single-letter data are allowed as background. At depth four, (10.4) provides the first exact transition between these two questions. A general algorithm should track lower-length merged words, not just the top-length Lyndon terms. Its output may be a filtered module presentation rather than the vector-space count  $N_d$ .

**3. Adding convergent shuffle and functional equations.** The single-letter-closed locus in Theorem 5.2 is complete for the stipulated shuffle vocabulary. Determine its enlargement when specific convergent iterated-integral shuffle identities, reflection identities, or functional equations are added. The target is a new exact presentation and a replayable certificate, not an inference from failed integer-relation searches.

**4. Colored commutators and conductors.** Extend the decorated alphabet by root-of-unity colors. Merging then multiplies colors while adding both exponents. Derive the corresponding contact terms, character projectors, and ordinary Mellin formulas, including what changes when the color is nontrivial and the one-variable pole is absent. The rational-shift formulas of Proposition 8.2 provide concrete consistency checks.

**5. Nonlinear and tangential finite parts.** For a fixed analytic curve through the all-one point, the required jet orders depend on its orders of contact with the prefix divisors. Apply the orbit-sum theorem to the resulting finite set of bidegrees and derive an exact orientation count in terms of those contact orders. Unlike the transverse formula, the set need not lie on the single diagonal  $j + r = d$ . A coefficient-identifiability theorem must then account for constraints among the curve's Taylor coefficients.

**6. Eliminating higher ordered terms from the Gamma integrals.** Formula (10.11) gives an ordinary integral for every  $\gamma_q$ . Seek additional exact identities that eliminate  $D_q$  in favor of lower-depth functions or classical polylogarithm order derivatives. It is essential to specify which function class is permitted; simply replacing  $D_q$  by the derivative in (10.10) would only restate the fundamental theorem of calculus.

**7. Rational ray designs at higher depth.** Theorem 5.1 proves existence of rational positive sample rays. Find explicit structured families with transparent inverse matrices for  $d = 5, 6$ , using the nine and twenty-one orientation coordinates, respectively. The aim is symbolic recoverability and simple exact coefficients, not a numerical conditioning claim without an accompanying analysis.

**8. Formal verification of the algebraic–analytic interface.** A first proof-assistant target is the finite word algebra: merge, quasi-shuffle, partition-lattice symmetrization, the orbit quotient, and the ray matrix. The analytic target can then isolate normal convergence of the outer-tail series, the uniqueness of the cutoff polynomial, and parameter differentiation. This separation avoids treating a successful finite symbolic script as a formal proof of an analytic identity.

## A The numerical harmonic constant-term evaluator

The evaluator represents a truncated asymptotic expansion in  $x = N + a$  as a dictionary of coefficients of  $x^{-p} \log^q x$ . Finite nested sums are evaluated by the exact ordering recurrence

$$W_{(k,m)v}(n+1) = W_{(k,m)v}(n) + (n+a)^{-k} \log^m(n+a) W_v(n), \quad W_\emptyset = 1.$$

For a prospective increment expansion  $f(x)$ , a formal discrete primitive is

$$\mathfrak{E}f(x) = \int f(x) dx - \frac{1}{2}f(x) + \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} f^{(2r-1)}(x). \quad (\text{A.1})$$

Every integration constant is set to zero in the logarithmic polynomial. For a monomial, the primitives are

$$\begin{aligned} \int x^{-1} \log^q x dx &= \frac{\log^{q+1} x}{q+1}, \\ \int x^{-p} \log^q x dx &= -x^{1-p} \sum_{h=0}^q \frac{q! \log^{q-h} x}{(q-h)!(p-1)^{h+1}}, \quad p > 1. \end{aligned}$$

All derivatives and products of the finite expansions are elementary. The constant is estimated recursively as the finite sum at  $N$  minus its nonconstant asymptotic primitive at  $N + a$ . Convergent outer sums use the same primitive to approximate their tails. For the commutator series, exactly cancelling nonsummable polynomial terms are removed symbolically by their exponent pattern before the numerical tail is formed.

This algorithm uses an *asymptotic* expansion. Increasing its order indefinitely at fixed cutoff is not asserted to converge. The recorded tests increase both cutoff and order, compare with separate single-variable calls, and retain all residuals. No interval enclosure or optimal-truncation bound is implemented. The mathematical proofs in the body of the article do not depend on the numerical evaluator.

## B Finite normal-form and ray-recovery algorithms

For a given depth  $d$ , enumerate all  $j = 1, \dots, d$  and weak compositions  $\mathbf{m}$  of  $d - j$  into  $j$  parts. Group the tuples by their sorted version. For each group, generate the labelled symmetric-sum certificate from (4.1), divide by the repeated-entry multiplicity, and apply (4.5). Substitute the monomial coefficient from (3.7). All operations are finite rational additions, multiplications, and coefficient collections.

For a practical positive-ray search, enumerate positive integer vectors in a fixed order, omit duplicates under common rescaling, and append a row whenever exact rational elimination increases the orientation-matrix rank. Theorem 5.1 proves that some finite stage attains rank  $N_d$ : rational positive rays can be scaled to positive integer ones, and a common slope scaling leaves (3.7) unchanged. This is a termination theorem for a specified algebraic enumeration. It is not a claim that the first full-rank matrix has small entries or is numerically well conditioned.

The certificate checker expands quasi-shuffle products recursively via

$$(av) * (bw) = a(v * (bw)) + b((av) * w) + (a \circ b)(v * w).$$

The empty word is the multiplicative unit. Word coefficients are integers, so a vanishing residual is checked by exact coefficient cancellation. The explicit  $3 \times 3$  matrix in (9.9) and the formal Gamma bridges are examples of certificates produced without numerical constants of any kind.

## C A compact theorem dependency map

The analytic foundation consists of the harmonic constant-term character, the outer-tail Mellin representation, and the compatible germ. The orientation rank uses only the universal quasi-shuffle algebra and the partition-lattice symmetric sum; it does not require analytic continuation. Ray identifiability uses only rational functions of the slopes. The ray finite-part interpretation then joins these two foundations.

The commutator series is obtained from the depth-two outer-tail recursion. The quartic formula is a finite substitution into the ray coefficient formula. The Gamma bridge uses the parameter derivative of words and three finite stuffle rows. The curvature identity uses the depth-three outer-tail recursion. Thus none of the new analytic identities is justified by the numerical regression table, and the exact rank theorem does not assume any unproved arithmetic property of zeta or Stieltjes values.

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